# VoxShield Architecture – Beginner to Advanced Guide

## 1. High-Level Overview

VoxShield is an audio deepfake detection model. Its job is to take a speech recording and decide whether it is likely genuine human speech or a synthetic/converted deepfake. To do this reliably, the model looks at the same audio in two different ways and then combines what it learns using attention-based layers.

The architecture has four main parts: (1) input and preprocessing, (2) a self-supervised learning (SSL) branch based on wav2vec2, (3) a spectro-temporal CNN branch based on LFCC/Log-Mel spectrograms, and (4) fusion, attention, and classification layers that produce a calibrated real/fake score.

## 2. Input Audio and Preprocessing

Input: VoxShield expects a mono speech waveform, typically resampled to 16 kHz. This is a simple one-dimensional array of audio samples.

Normalization: The waveform is normalized so that its amplitude stays in a stable range. This helps the model focus on patterns, not absolute volume.

Segmentation: Long recordings are split into fixed-length windows (for example, 4-second segments with some overlap). Each segment is processed separately and later combined. This keeps the computation manageable and ensures that the model sees local details.

## 3. Self-Supervised wav2vec2 Branch

The first main branch feeds the raw waveform into a pre-trained self-supervised encoder such as wav2vec2. Self-supervised learning means the model has already been trained on a large amount of unlabeled speech to learn generic speech representations.

wav2vec2 Encoder: This encoder outputs a sequence of high-level feature vectors. Each vector summarizes a short time slice of the audio, capturing information about phonemes, pronunciation, and acoustic texture that deepfake systems may not perfectly replicate.

Projection Layer: The wav2vec2 features usually have a high dimension (for example, 768). A linear projection layer reduces this to a smaller dimension (for example, 128). This makes the later fusion step more efficient and easier to train.

## 4. Spectro-Temporal CNN Branch

The second branch looks at the same audio in the time–frequency domain. Instead of raw samples, we compute a spectrogram-like representation such as LFCC (Linear-Frequency Cepstral Coefficients) or Log-Mel spectrograms.

LFCC / Log-Mel: These features describe how energy is distributed across frequencies over time. Deepfake generators often leave small but consistent artifacts in certain frequency bands, which these features make easier to detect.

2D CNN Front-End: The spectrogram is treated like an image (frequency × time) and passed through a small 2D convolutional neural network. Typically this includes two Conv2D + BatchNorm + ReLU blocks with pooling. The CNN learns local spectro-temporal patterns such as unnatural harmonics or noise textures.

Flattening and Projection: After the CNN, the frequency dimension is collapsed, leaving a time sequence of feature vectors. Another projection layer maps these to the same feature size used in the wav2vec2 branch so both branches can be fused.

## 5. Feature Fusion Layer

At this point, we have two aligned sequences of features for each segment: one from the wav2vec2 branch and one from the spectro-temporal CNN branch. VoxShield combines them in a fusion layer.

Concatenation: For each time step, the corresponding wav2vec2 feature and spectrogram feature are concatenated into a single vector. This directly joins high-level SSL information with low-level spectral cues.

Fusion Projection: A linear fusion layer then projects this concatenated vector back down to a moderate size. This keeps the model compact and encourages it to learn a unified representation that mixes both views of the audio.

## 6. Temporal Attention / Transformer Block

The fused sequence is passed to a temporal attention block, implemented as one or more Transformer encoder layers. A Transformer uses self-attention to let each time step look at other time steps in the segment.

Self-Attention: Self-attention computes how strongly each frame should attend to every other frame. This helps the model detect patterns such as repeated artifacts, inconsistencies in prosody, or sudden changes that are typical of deepfake audio.

Positional Encoding: Because attention itself is order-agnostic, positional encodings are added so the model knows the relative position of each frame in time.

## 7. Attention Pooling and Clip Embedding

After the Transformer block, VoxShield needs to summarize the whole segment into a single embedding vector. Instead of simple averaging, it uses attention pooling.

Attention Pooling: A small attention mechanism learns a weight for each time step. Frames that look more suspicious (or more informative) get higher weights. The final segment embedding is a weighted sum of all frames.

## 8. Classification Head

The pooled embedding passes through a small feed-forward network, typically one or two dense layers with non-linear activation and dropout to prevent overfitting.

Output Layer: The final layer outputs a single logit or a two-dimensional vector. After applying a sigmoid or softmax, we obtain a probability that the segment is fake, and by complement, the probability that it is real.

## 9. Calibration and Risk Scoring

Raw probabilities from neural networks are often not perfectly calibrated. VoxShield can use a simple calibration step, such as temperature scaling or Platt scaling, on a validation set so that predicted probabilities better match actual error rates.

Risk Bands: For practical use, segment or clip scores can be mapped into bands such as Low Risk (likely genuine), Medium Risk (uncertain), and High Risk (likely fake). These bands are determined by thresholds chosen from validation data.

## 10. End-to-End Training Flow

Training Data: The model is trained on labeled audio from benchmarks like ASVspoof (genuine vs spoofed speech). Each segment is paired with a binary label.

Loss Function: VoxShield uses binary cross-entropy loss on the predicted fake probabilities. Optionally, segments from the same original recording can be encouraged to have consistent scores using an additional consistency loss.

Optimization Strategy: In practice, wav2vec2 layers are often frozen at first while the CNN branch, fusion layer, Transformer, and classifier are trained. Later, the last few layers of wav2vec2 can be fine-tuned with a lower learning rate to squeeze out extra performance.

By combining self-supervised waveform features, spectrogram-based CNN features, attention-based fusion, and calibrated classification, VoxShield provides a modern and robust architecture for detecting audio deepfakes.